MODULE 28 · DOMAIN 4 - IMPLEMENT TEXT ANALYSIS SOLUTIONS

Speech for agents

Voice Live and turn detection, speech to text options, custom speech and phrase lists, text to speech and SSML, audio as a reasoning input, and speech translation.

DOMAIN 4 - Text analysis WEIGHT 10-15% MODULE 28 of 39

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A voice agent is a loop: **hear**, **think**, **speak**. Build each stage yourself, or hand the whole loop to **one service**.

Three families of speech technology

FAMILY	WHAT IT DOES	CHOOSE IT FOR
Azure Speech	Speech to text, text to speech, translation, custom models	Individual stages, customisation, containers
Voice Live API	Speech-to-speech conversation in one API	Voice agents with natural turn-taking
Azure OpenAI audio models	Realtime API and audio in model calls	Reasoning directly over audio

ON THE EXAM

Microsoft's guidance: use Azure OpenAI audio when audio and reasoning belong in a single model call.

Build the pipeline, or use speech-to-speech?

CASCADE PIPELINE

Separate stages

- Speech to text, then a model, then text to speech
- Customise and inspect every stage
- You handle turn-taking and interruptions
- More moving parts, more latency to manage

SPEECH-TO-SPEECH

One conversational loop

- Voice Live, or the Realtime API
- Turn detection and interruptions built in
- Lowest latency for live conversation
- Less control over each individual stage

THE SIGNAL

'Natural conversation with interruptions, without stitching services together' points to speech-to-speech.

What Voice Live handles for you

- **Speech recognition, a generative model and text to speech**, combined in one API.
- Model choice from **GPT-Realtime** to GPT-5, GPT-4.1 and Phi, with no model deployment to manage.
- **Noise suppression**, **echo cancellation**, and **interruption detection**.
- **Avatars**, and **function calling** for actions and grounded answers, the VoiceRAG pattern.
- Compatible with **Realtime API events**, over WebSocket from your back end.

> ON THE EXAM
>
> Echo cancellation stops the agent hearing its own voice and interrupting itself.

Voice Live with Foundry Agent Service

- Connect to an **agent** instead of a model: pass **agent-name** and **agent-project-name**.
- The agent brings its **tools, knowledge, memory and governance** into the voice session.
- Voice Live for **prompt agents** is **generally available**, since Build 2026.
- Voice Live with **hosted agents** is in **public preview**.
- The same agent can serve text chat and voice, without rebuilding its logic.

> ON THE EXAM
>
> An existing Foundry agent becomes a voice agent through Voice Live, not by adding speech tools to its instructions.

Knowing when the user has finished

OPTION	HOW IT DECIDES	NOTES
Azure semantic VAD	A semantic speech model	The default; robust in noisy places
Azure end-of-utterance	Detects the end of a thought	Allows natural pauses mid-sentence
OpenAI semantic VAD	A model judges completion	Only with gpt-realtime models

THE SIGNAL

An agent that cuts people off while they pause to think needs end-of-utterance detection.

Four ways to transcribe

OPTION	HOW IT WORKS	CHOOSE IT FOR
Real-time	Streams results as people speak	Live captions, voice input to agents
Fast transcription	Synchronous, faster than real time, per file	A whole recording, as quickly as possible
Batch transcription	Asynchronous, large volumes of files	Call-centre archives, with diarization
LLM speech (preview)	LLM-enhanced transcription and translation	Higher quality, with prompt tuning

THE TIEBREAKER

Live conversation needs real-time. A file you need back quickly: fast. Thousands of files overnight: batch.

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Phrase list or custom speech?

PHRASE LIST

A runtime hint

- Pass names and terms with the request
- No training at all
- Works with real-time and fast transcription
- Update instantly as terms change

CUSTOM SPEECH

A trained model

- Train on your text, or audio with transcripts
- Handles vocabulary and acoustic conditions
- Deploy to a custom endpoint for real-time
- Retrain when data changes

ON THE EXAM

New product names this week, no audio data: phrase list. Noisy factory floors and accents: custom speech with audio.

Training data for custom speech

- **Plain text** teaches domain vocabulary. Every base model supports it.
- **Audio with human-labelled transcripts** teaches acoustic conditions: noise, accents, microphones.
- Some locales also accept **structured text** and **pronunciation** data.
- Real-time use needs a **custom endpoint**. **Batch transcription doesn't**, which saves cost.
- Compare accuracy against the base model before switching production traffic.

> ON THE EXAM
>
> Text fixes 'what words'. Audio fixes 'how it sounds'. Match the data to the failure.

Voices for agents

- **Standard neural voices** at 24 kHz and 48 kHz, across many languages.
- **HD voices** for more expressive speech, and **Azure OpenAI voices** in preview, in two regions.
- **Multilingual voices** detect the language of the input text automatically.
- **SSML** controls pitch, pauses, pronunciation, speaking rate and volume.
- Billing counts **every character**, including spaces and punctuation.

> ON THE EXAM
>
> Cost follows the length of what the agent says, so concise replies are cheaper to speak.

SSML for a clearer agent reply

```
<speak version="1.0" xmlns="http://www.w3.org/2001/10/synthesis" xml:lang="en-GB">
  <voice name="en-GB-SoniaNeural">
    Your parcel arrives on
    <say-as interpret-as="date" format="dmy">18/09/2026</say-as>.
    <break time="400ms"/>
    <prosody rate="-10%">Your reference is
      <say-as interpret-as="characters">AB12</say-as>.</prosody>
  </voice>
</speak>
```

ON THE EXAM

say-as reads the date correctly and the reference letter by letter. prosody slows down the part people must write down.

Custom voices are Limited Access

- **Custom voice** covers **professional voice fine-tuning** and **personal voice**.
- Custom neural voice, personal voice and custom avatars are **Limited Access**.
- Registration and **approved use cases** are required, with voice talent consent.
- **Prebuilt** voices need no registration for use cases within the acceptable use policy.
- Tell users they're hearing a synthetic voice, especially one modelled on a real person.

> ON THE EXAM
>
> 'Clone our CEO's voice by tomorrow' isn't a configuration task. It needs Limited Access approval first.

Reasoning over audio directly

- Audio-capable models, like **gpt-audio**, accept audio in **Chat Completions** calls.
- The **Realtime API** runs voice-agent, **translation** and **transcription** sessions.
- Realtime sessions last up to **60 minutes**, with audio sent in small base64 chunks.
- Realtime quotas are **separate** from chat completions quotas.
- **GPT Realtime 2.0**, in preview, adds reasoning to live voice conversation.

> ON THE EXAM
>
> Access needs the Cognitive Services OpenAI User role, as with other Azure OpenAI calls.

A transcript keeps the **words** and loses the **voice**. If **how** something was said matters, reason over the audio.

Standard speech translation or Live Interpreter?

STANDARD SPEECH TRANSLATION

Known input language

- Set the input language
- Up to two target languages per call
- Text translations, with synthesis available

LIVE INTERPRETER

Any language, any time

- No input language needed
- Handles speakers switching languages mid-session
- Natural voice that keeps the speaker's style
- Can use a personal voice

THE TIEBREAKER

Speakers who switch languages unpredictably need Live Interpreter. A fixed input language is the wrong assumption.

More languages, and model-based options

- One call translates into **two target languages** directly.
- More than two needs a **Foundry resource**, or separate translation, with fees for extra languages.
- The Realtime API's **translation session** and **gpt-realtime-translate** stream translation.
- A cascade still works: **transcribe**, translate with **Translator** from module 27, then **synthesise**.
- Live Interpreter currently transcribes the **target** language only, not the source.

> ON THE EXAM
>
> The cascade gives you tone and gender controls from Translator's LLM option at the middle step.

Requirement to approach

REQUIREMENT	APPROACH
Give an existing Foundry agent a natural voice	Voice Live, connected by agent name
Agent interrupts people mid-pause	End-of-utterance turn detection
This week's product names in live calls	Phrase list
Accents and noise on a factory floor	Custom speech with audio and transcripts
Read dates and references clearly	SSML say-as and prosody
Did the customer sound hesitant?	An audio model reasoning over the recording
Speakers switch languages mid-talk	Live Interpreter

ON THE EXAM

Hear, think, speak: find which stage the requirement belongs to, then choose within that stage.

A bank has a Foundry prompt agent that already handles account questions in text chat, with tools and knowledge configured.

It now wants callers to talk to the same agent naturally, with interruptions handled and background noise suppressed, without stitching services together.

What should the team use?

A Real-time speech to text, the agent, then text to speech, wired up.

B The Voice Live API, connected to the existing agent by agent name.

C Batch transcription of each call, followed by a text-to-speech reply.

D Live Interpreter, translating each caller's speech for the agent.

◆ PAUSE AND DECIDE

What should the team use?

A. A hand-built cascade.
Works, but it's the stitching the bank wants to avoid, and interruptions are left to them.

B. Voice Live with the agent.
One API, with turn-taking, interruptions and noise suppression, reusing the agent as is.

C. Batch transcription.
Asynchronous, for recorded files. Useless for a live conversation.

D. Live Interpreter.
Translates between languages. Nobody asked for translation.

THE SIGNAL

An existing agent, natural live voice, no stitching: Voice Live, connected by agent name.

> A retailer's voice assistant uses real-time speech to text. It keeps mistranscribing new product names, which change every week.
>
> The team has no recorded audio of the new names, and needs them recognised from Monday.

What should the team do?

A Train custom speech with audio and human-labelled transcripts each week.

B Switch from real-time transcription to batch transcription for accuracy.

C Send the week's product names as a phrase list with each recognition.

D Add the product names to an SSML lexicon for the assistant's voice.

◆ PAUSE AND DECIDE

What should the team do?

A Weekly custom speech with audio.

There's no audio to train on, and weekly retraining can't beat Monday.

B Batch transcription.

Asynchronous, so the live assistant stops working in real time.

C A phrase list.

A runtime hint with no training, updated instantly each week.

D SSML lexicon.

Controls how the assistant pronounces words, not what it recognises.

THE SIGNAL

Changing names, no audio, needed immediately: a phrase list, which needs no training.

A sales team reviews recorded calls. Managers want to know whether customers sounded hesitant or confident when discussing price.

A pilot that transcribed calls and ran text sentiment rated most hesitant customers as neutral.

What should the team change?

A Send the call audio to an audio-capable model to reason over voice.

B Switch the transcription to custom speech trained on sales vocabulary.

C Run a second text sentiment model on the same transcripts and average.

D Convert transcripts back into speech with an expressive HD voice first.

◆ PAUSE AND DECIDE

What should the team change?

A Audio model over the voice.

Hesitation lives in pauses and delivery, which only the audio contains.

B Custom speech.

Better words in the transcript, but still no voice.

C Average two text models.

Both read the same words, and both miss the delivery.

D Resynthesise with HD voice.

A new voice reading the text can't restore how the customer sounded.

THE SIGNAL

'How they sounded' can't survive transcription. Reason over the audio itself.

> An international conference livestreams talks. Several speakers switch between English and Spanish in the middle of their talks.
>
> Remote attendees need to hear each talk translated, in a natural voice, as it happens.

What should the organisers use?

A Real-time speech to text set to en-US, then Translator on the text.

B Standard speech translation, with the input language set to English.

C Batch transcription after each talk, then translated subtitles.

D Live Interpreter, which identifies and follows the spoken language.

◆ PAUSE AND DECIDE

What should the organisers use?

A STT fixed to en-US.

Spanish passages are transcribed as garbled English, and there's no spoken output.

B Speech translation fixed to English.

The same fixed-language problem whenever a speaker switches.

C Batch after each talk.

Not live, and subtitles aren't spoken translation.

D Live Interpreter.

Follows language switches and speaks the translation naturally, in real time.

THE TIEBREAKER

Unpredictable language switching plus live spoken output: Live Interpreter.

Speech for agents in five lines

- **Voice Live** runs the hear-think-speak loop, and connects to Foundry agents by name.
- **Real-time** for live, **fast** for one file quickly, **batch** for volume.
- **Phrase list** for changing terms now; **custom speech** for vocabulary and acoustics.
- **SSML** shapes speech; custom and personal voices are **Limited Access**.
- **How it was said** needs audio reasoning. **Switching languages** needs Live Interpreter.

ON THE EXAM

Domain 4 is complete. Next, module 29 begins Domain 5: ingesting and indexing content for grounding.